AIRCRAFT DESIGN SUMMARY

Max take-off mass (MTOM)	**71,062**	kg
Operating empty mass	**35,727**	kg
Mission fuel (design mission)	**21,085**	kg
Design payload	**14,250**	kg
Design wing loading W/S	**5,281**	N/m2
Design wing loading m/S	**538.5**	kg/m2
Design thrust-to-weight T/W (SL static)	**0.273**	-
Critical constraint	**Take-off + cruise**	
Wing area S	**132.0**	m2
Wing span b	**35.41**	m
Mean aerodynamic chord	**4.17**	m
Quarter-chord sweep	**26.2**	deg
Total SL static thrust	**190.1**	kN
Thrust per engine	**95.0**	kN
Cruise L/D	**17.46**	-
Fuselage length	**35.62**	m
Fuselage outer diameter	**3.98**	m
Seats abreast x rows	**6 x 25**	
Horizontal tail area	**34.4**	m2
Vertical tail area	**26.2**	m2
Main gear: struts x wheels per strut	**2 x 2**	
Main tyre diameter x width	**1.15 x 0.42**	m
Range with maximum payload	**3,590**	km
Take-off field length achieved	**2,200**	m
Landing field length achieved	**1,584**	m
Service ceiling	**11,259**	m
Time to climb to cruise altitude	**18.8**	min
Design limit load factor	**2.50**	-
Static margin at MTOM	**11.3%**	MAC
CDo from drag build-up	**0.0193**	-
Empty mass from component build-up	**35,845**	kg
Cost per seat-kilometre	**4.60**	US cents
Closest real aircraft	**Airbus A320-200**	
Balanced field length (Raymer)	**2,247**	m
Best initial cruise altitude	**11,259**	m
CO2 per seat-kilometre	**72.5**	g
Main tyre pressure	**1.08**	MPa
Requirements met	**14 of 14**	

CHECKS

Weight iteration	**OK - converged**
Max payload >= design payload	**OK**
Payload per passenger plausible	**OK**
Range with maximum payload	**OK**
Wing fuel volume	**OK**
Gear tip-back angle >= 15 deg	**OK**
Engine ground clearance	**OK**
Thrust lapse / TSFC unit plausibility	**OK**
Take-off distance	**OK - 0 m margin on TODA**
Landing distance	**OK - 86 m margin on LDA**
Service ceiling	**OK - 591 m above cruise altitude**
Static stability	**OK**
Gear position vs calculated CG	**OK**
CDo input vs build-up	**OK - within 20 % of the build-up**
Cargo hold volume	**OK**
Climb gradients (engine out)	**OK - every climb requirement is met**
Balanced field length cross-check	**NOTE - Raymer BFL is 47 m over TODA (the sizing used the Loftin correlation)**
Trim on approach	**OK**
Take-off rotation	**OK**
Engine-failure control	**OK**
Crosswind control	**OK**
Seats drawn	**OK**
Tailplane position	**OK**
Tyre pressure and pavement loading	**OK for major airports (PCN 60-80)**

REQUIREMENT	Required / Achieved	Status
Aspect ratio	9.500 / 9.500	OK
Oswald efficiency factor	0.800 / 0.800	Used as given
Zero-lift drag coefficient	0.018 / 0.019	OK
Maximum payload (kg)	17,500 / 17,500	OK
Take-off distance available (m)	2,200 / 2,200	OK
Landing distance available (m)	1,670 / 1584.410	OK
CLmax, take-off	2.100 / 2.100	OK
CLmax, landing	2.700 / 2.700	OK
Cruise thrust lapse	0.220 / 0.198	OK
Cruise Mach number	0.780 / 0.820	OK
Cruise TSFC	0.600 / 0.600	Used as given
Typical passengers	150 / 150	OK
Design range	5,000 / 5,000	OK
Typical passenger payload (kg)	14,250 / 14,250	OK











